Yukang Shen

Embodied AI Researcher · Software Engineer
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"The physical world never stands still, neither should intelligence."

RESEARCH INTERESTS

I investigate continual learning and memory systems for embodied agents, focusing on how physical robots retain experience, reduce false-identity updates, and adapt under resource constraints. Related interests include multimodal perception, sim-to-real transfer, and robotic manipulation.

EDUCATION

Kennesaw State University, M.S. in Software Engineering **Jan. 2025 – Nov. 2026 (expected)**
GPA: 3.77/4.00

Heilongjiang University, B.E. in Computer Network Engineering **Sep. 2016 – Jul. 2020**
GPA: 8.129/10.0

HONORS

- **Kennesaw State University**: Second Place, KSU C-Day Computing Showcase, Fall 2025
- **Heilongjiang University**: National Encouragement Scholarship; First-Class Scholarship, 2018, 2019

PROFESSIONAL EXPERIENCE

City University of Hong Kong, Research Assistant(Product Developer) **Nov. 2024 – Jan. 2025**

- Researched Snowflake and comparable data platforms and use cases, then worked with the team to align its research on federated learning and encrypted data transactions with product requirements.
- Developed the cross-institutional data marketplace, including access-control and transaction-audit workflows.

SenseTime Group Limited, Software Engineer **Nov. 2020 – Sep. 2023**

- Delivered numerous internal and commercial products; several served **5,000+** users.
- Built a range of internal enterprise systems, including recruitment/onboarding systems, a company-wide office automation (OA) portal, and business intelligence (BI) platforms, interactive annual-report applications, and smart meeting-room booking and screen-casting systems across desktop, web, mobile, and integrated hardware.
- Mentored 5 interns; named the **department's best employee of the year in 2022**.

RESEARCH EXPERIENCE

Similarity Is Not Identity: Evidence-Gated Writes for Embodied Memory **2026 – Present**
In progress
Question: Can continuity-gated writes reduce wrong-object updates among visually similar objects?

- Modeled object identity tracking as a memory write-verification problem to reduce false updates among visually similar objects.
- Developed a multi-stage filtering pipeline combining geometric constraints and visual feature similarity, enforcing strict continuity checks (e.g., uninterrupted tracking) before updating object records.

A Real-Calibrated Synthetic-First Data Engine **2025 – 2026**
Independent research; sole-author preprint
Question: Can synthetic data carry a vision model where labeled real data are scarce? Side-view, shoulder-exposed images are rare, creating a long-tail gap for deltoid-region segmentation.

- Developed a Python-based synthetic data engine combining diffusion-based image generation, SAM-based semantic filtering, and automated dataset packaging.
- Evaluated YOLOv11-pose across 5 training configurations on a 280-image real test set, improving pose mAP@0.5:0.95 from 0.389 to 0.411 (+5.7% relative) with hybrid training.

PUBLICATIONS & PATENTS

- *6G-enabled Edge AI for Metaverse: Challenges, Methods, and Future Research Directions.* Luyi Chang, Zhe Zhang, Pei Li, Shan Xi, Wei Guo, **Yukang Shen**, Zehui Xiong, Jiawen Kang, Dusit Niyato, Xiuquan Qiao, Yi Wu. *Journal of Communications and Information Networks*, 2022.
- *A Real-Calibrated Synthetic-First Data Engine.* **Yukang Shen (sole author)**. Preprint, arXiv:2605.09699, 2026.
- *Job Entry Guiding Method and Device, Electronic Equipment, Storage Medium and Program Product.* Patent No. CN114581057.

PRESENTATIONS

- *MemoryEIL: An Enhanced Memory Layer Architecture for Heterogeneous Robots.* KSU C-Day Computing Showcase, Spring 2026.
- *A Synthetic Data Engine for Explainable Injection-Area Perception.* KSU C-Day Computing Showcase, Fall 2025.
- *XR Agent: An MLLM-Powered Extended-Reality System on Meta Quest 3.* KSU C-Day Computing Showcase, Spring 2025.

SKILLS

AI Research: Python, PyTorch, ROS, AI2-THOR, Isaac Sim, MuJoCo, LIBERO; continual and reinforcement learning, VLA models, embodied memory and retrieval, multimodal perception, synthetic data, model evaluation, sim-to-real.

Product Development: TypeScript, React, Python, Node.js, CI/CD; requirements engineering, project management.